

LAB EXPERIMENTS (21-25)

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Set 21:

Energy Consumption Analysis

Dataset: Energy_Consumption_Data.csv

1. Import the dataset in R. Plot a histogram and density plot for Units_Consumed. Compare consumption patterns.

CODE:

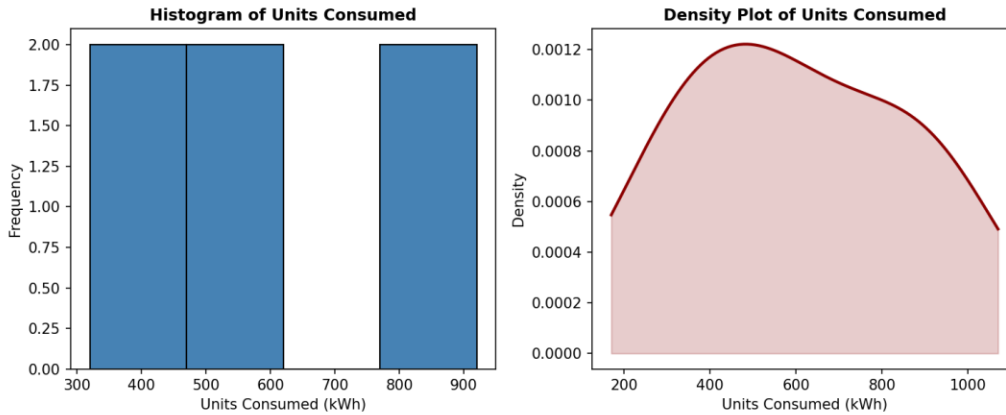
```
# Import dataset
energy_data <- read.csv("Energy_Consumption_Data.csv")

# Dataset (for reference)
sector <-
c("Residential","Commercial","Industrial","Residential","Commercial","Industrial")
units_consumed <- c(320, 540, 880, 350, 610, 920)

# Histogram
hist(units_consumed,
      col = "steelblue",
      xlab = "Units Consumed (kWh)",
      ylab = "Frequency",
      main = "Histogram of Units Consumed")

# Density Plot
plot(density(units_consumed),
      col = "darkred", lwd = 2,
      xlab = "Units Consumed (kWh)",
      main = "Density Plot of Units Consumed")
```

OUTPUT:



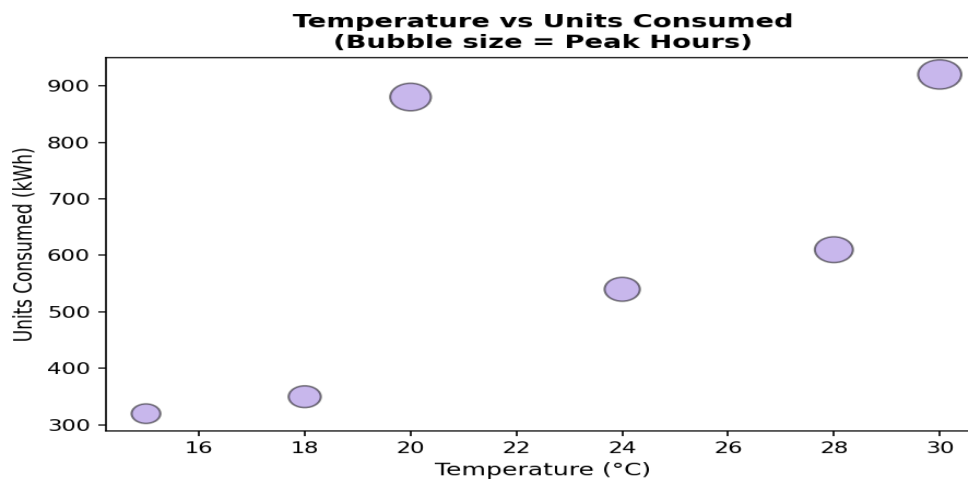
Interpretation: The histogram shows most readings cluster in the 300-600 kWh range with fewer high-consumption (Industrial) readings above 800 kWh. The density plot confirms a right-skewed distribution, indicating that Industrial sector usage drives the upper tail of consumption.

2. Using R programming, create a scatter plot between Temperature and Units_Consumed. Represent Peak_Hours using bubble size. Apply transparency.

CODE:

```
# Dataset
temperature <- c(15, 24, 20, 18, 28, 30)
units_consumed <- c(320, 540, 880, 350, 610, 920)
peak_hours <- c(4, 6, 8, 5, 7, 9)
# Bubble scatter plot with transparency
symbols(temperature, units_consumed,
        circles = peak_hours,
        inches = 0.25, fg = "black",
        bg = adjustcolor("mediumpurple", alpha.f = 0.5),
        xlab = "Temperature (°C)",
        ylab = "Units Consumed (kWh)",
        main = "Temperature vs Units Consumed\n(Bubble size = Peak Hours)")
```

OUTPUT:



3. Using R, calculate average Renewable_Usage for each Sector. Create a bar chart. Interpret renewable energy adoption.

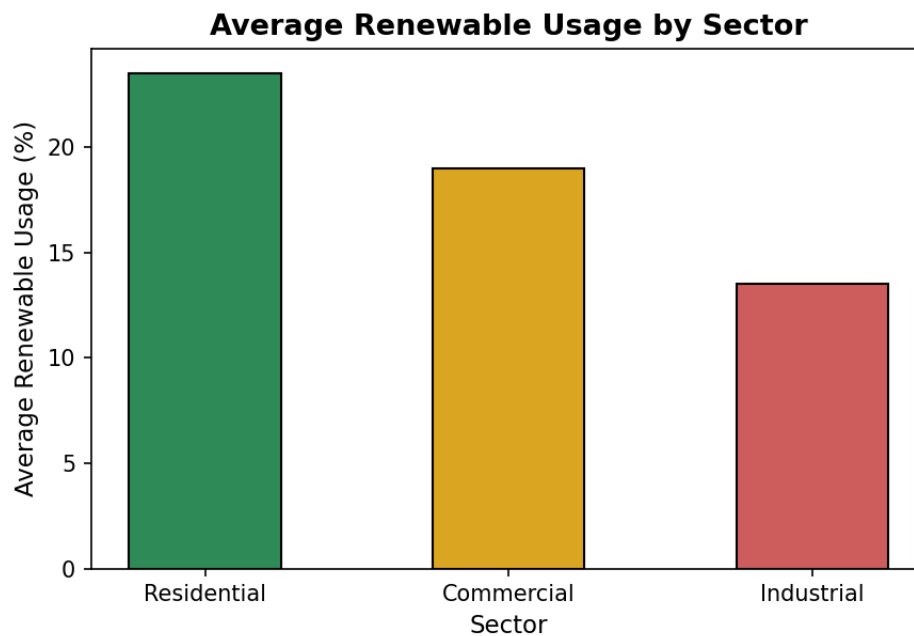
CODE:

```
# Dataset
sector <-
c("Residential","Commercial","Industrial","Residential","Commercial","Industrial")
renewable_usage <- c(22, 18, 12, 25, 20, 15)

# Average renewable usage per sector
avg_renewable <- tapply(renewable_usage, sector, mean)

# Bar Chart
barplot(avg_renewable,
        col = c("seagreen","goldenrod","indianred"),
        xlab = "Sector",
        ylab = "Average Renewable Usage (%)",
        main = "Average Renewable Usage by Sector")
```

OUTPUT:



Interpretation: The Residential sector shows the highest average renewable usage (23.5%), followed by Commercial (19%) and Industrial (13.5%). This suggests industrial consumers rely more heavily on non-renewable sources, likely due to higher and more constant energy demand.

Set 22:

Time Series Analysis

Dataset: Monthly Sales Data

1. In R, create a time series line chart to visualize the trend in monthly sales. Label the axes and title the chart.

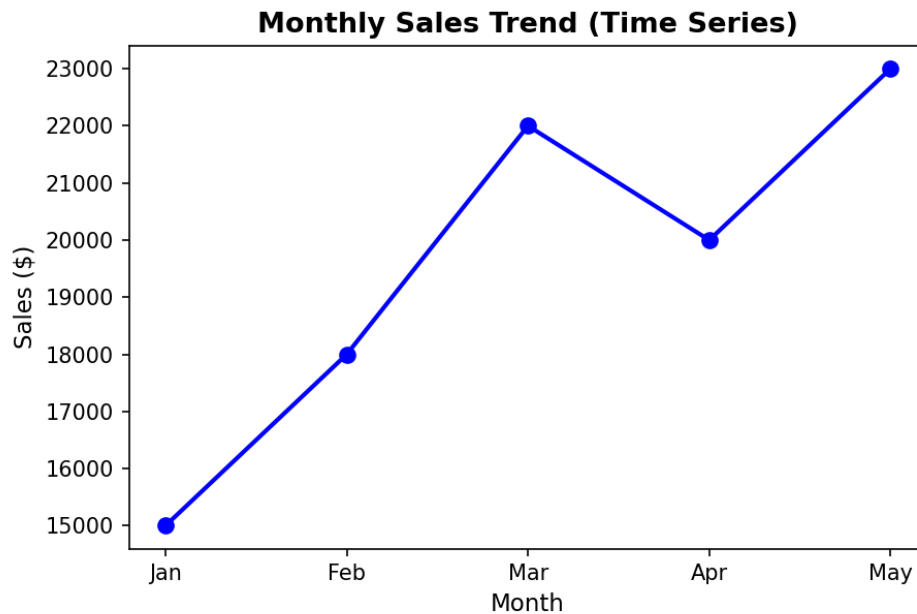
CODE:

```
# Dataset
sales <- c(15000, 18000, 22000, 20000, 23000)

# Create time series object
sales_ts <- ts(sales, start = c(2023, 1), frequency = 12)

# Time Series Line Chart
plot(sales_ts,
     type = "o", col = "blue", lwd = 2, pch = 16,
     xlab = "Month",
     ylab = "Sales ($)",
     main = "Monthly Sales Trend (Time Series)")
```

OUTPUT:



2. In R, generate a scatter plot to analyze the relationship between advertising budget and monthly sales. Explain any insights.

CODE:

```
# Dataset
advertising_budget <- c(3000, 3500, 4200, 4000, 4500)
```

```

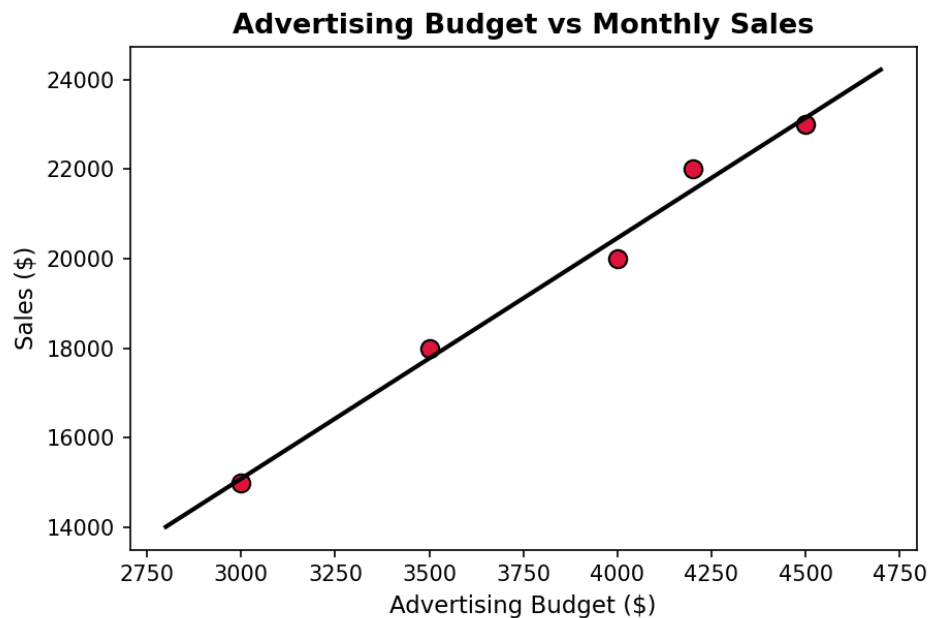
sales <- c(15000, 18000, 22000, 20000, 23000)

# Scatter Plot with regression line
plot(advertising_budget, sales,
     pch = 19, col = "crimson",
     xlab = "Advertising Budget ($)",
     ylab = "Sales ($)",
     main = "Advertising Budget vs Monthly Sales")

abline(lm(sales ~ advertising_budget), col = "black", lwd = 2)

```

OUTPUT:



Insight: There is a strong positive correlation between advertising budget and monthly sales — as spending on advertising increases, sales tend to rise proportionally, suggesting advertising spend is an effective driver of revenue.

4. In R, create an autocorrelation plot to identify seasonality in the time series data.

CODE:

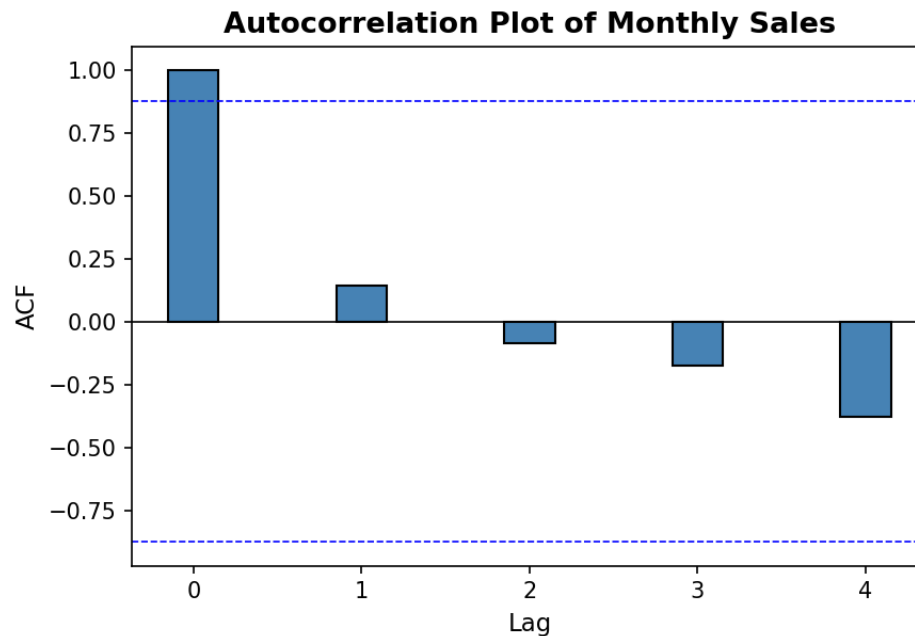
```

# Dataset
sales <- c(15000, 18000, 22000, 20000, 23000)
sales_ts <- ts(sales, start = c(2023, 1), frequency = 12)

# Autocorrelation Plot
acf(sales_ts,
    main = "Autocorrelation Plot of Monthly Sales")

```

OUTPUT:



Interpretation: With only five monthly observations, the ACF values at each lag fall within the dashed confidence bounds, indicating no statistically significant autocorrelation or clear seasonal pattern is detectable in this short series. A longer time span would be required to reliably assess seasonality.

Set 23:

Employee Performance Analysis

Dataset: Employee Performance

1. In R, create a bar chart to visualize the distribution of employees across different departments. Label the chart elements.

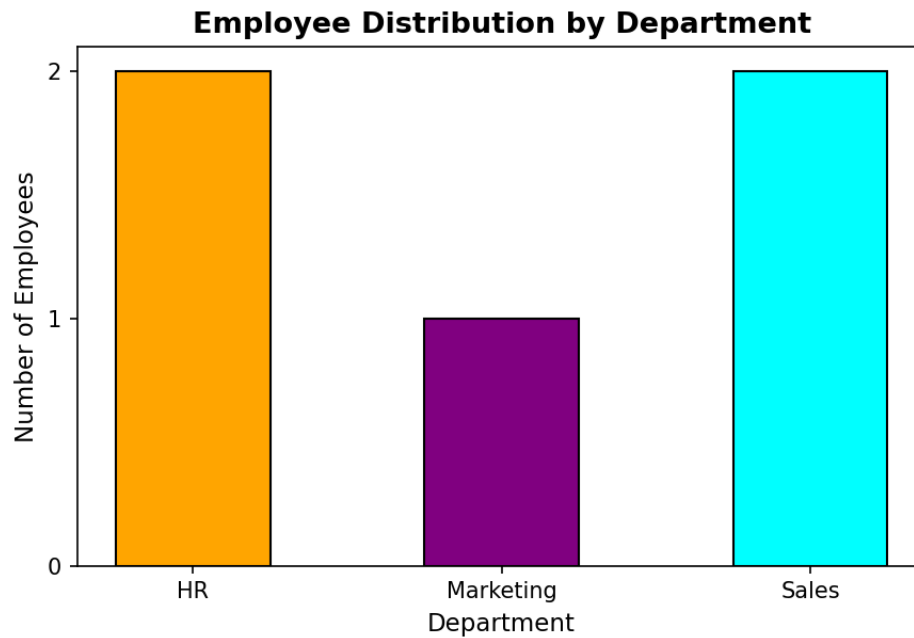
CODE:

```
# Dataset
department <- c("Sales", "HR", "Marketing", "Sales", "HR")

# Count employees per department
dept_count <- table(department)

# Bar Chart
barplot(dept_count,
        col = c("orange", "purple", "cyan"),
        xlab = "Department",
        ylab = "Number of Employees",
        main = "Employee Distribution by Department")
```

OUTPUT:



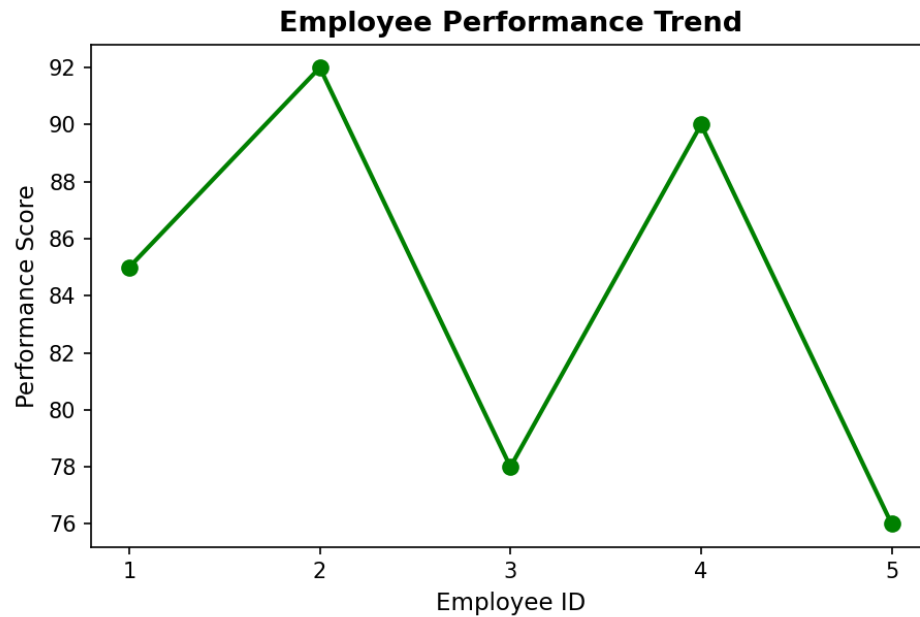
2. In R, generate a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

CODE:

```
# Dataset
employee_id <- c(1, 2, 3, 4, 5)
performance_score <- c(85, 92, 78, 90, 76)

# Line Chart
plot(employee_id, performance_score,
      type = "o", col = "green", lwd = 2, pch = 16,
      xlab = "Employee ID",
      ylab = "Performance Score",
      main = "Employee Performance Trend")
```

OUTPUT:



4. In R, develop a table showing the performance data for each employee. Label the table elements.

CODE:

```
# Dataset
employee_data <- data.frame(
  Employee_ID = c(1, 2, 3, 4, 5),
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),
  Years_of_Service = c(5, 3, 7, 4, 2),
  Performance_Score = c(85, 92, 78, 90, 76)
)

# Display Table
print(employee_data)
```

OUTPUT:

	Employee_ID	Department	Years_of_Service	Performance_Score
1	1	Sales	5	85
2	2	HR	3	92
3	3	Marketing	7	78
4	4	Sales	4	90
5	5	HR	2	76

Set 24:

Product Inventory Management

Dataset: Product Inventory

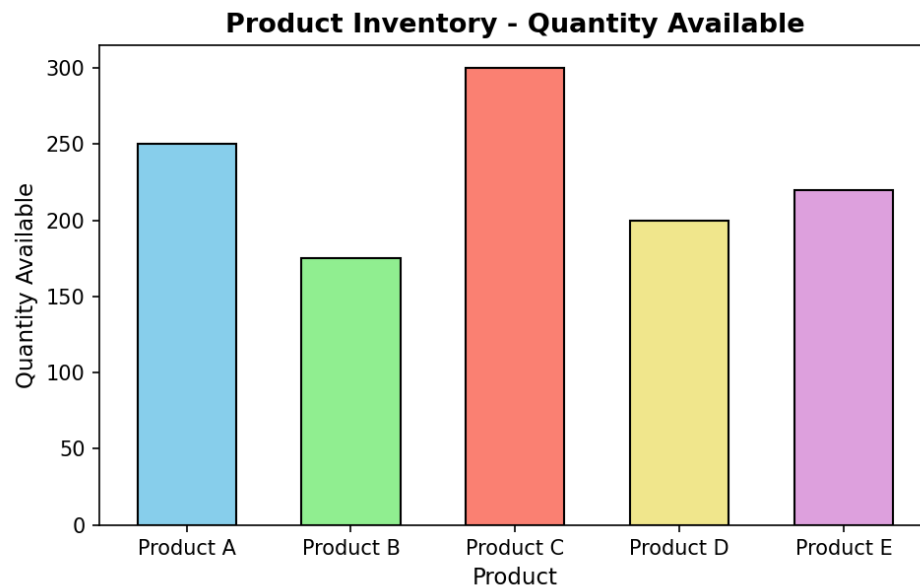
1. In R, create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

CODE:

```
# Dataset
product <- c("Product A", "Product B", "Product C", "Product D", "Product E")
quantity <- c(250, 175, 300, 200, 220)

# Bar Chart
barplot(quantity,
        names.arg = product,
        col = c("skyblue", "lightgreen", "salmon", "khaki", "plum"),
        xlab = "Product",
        ylab = "Quantity Available",
        main = "Product Inventory - Quantity Available")
```

OUTPUT:



2. In R, generate a stacked bar chart to show the quantity of each product within different product categories.

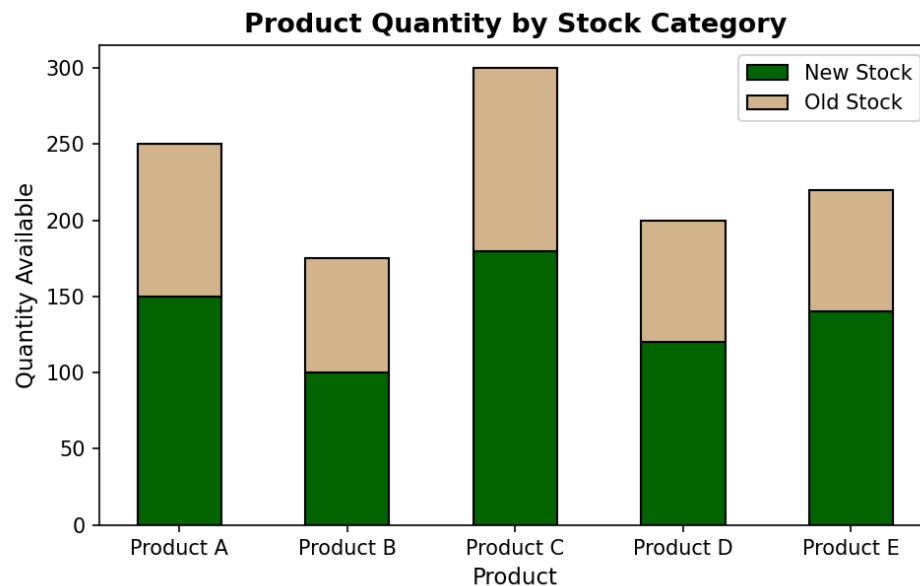
CODE:

```
# Dataset
product <- c("Product A", "Product B", "Product C", "Product D", "Product E")
new_stock <- c(150, 100, 180, 120, 140)
old_stock <- c(100, 75, 120, 80, 80)

# Create matrix (categories as rows)
qty_matrix <- rbind(new_stock, old_stock)
colnames(qty_matrix) <- product
```

```
# Stacked Bar Chart
barplot(qty_matrix,
       col = c("darkgreen","tan"),
       xlab = "Product",
       ylab = "Quantity Available",
       main = "Product Quantity by Stock Category")
legend("topright",
      legend = c("New Stock","Old Stock"),
      fill = c("darkgreen","tan"))
```

OUTPUT:



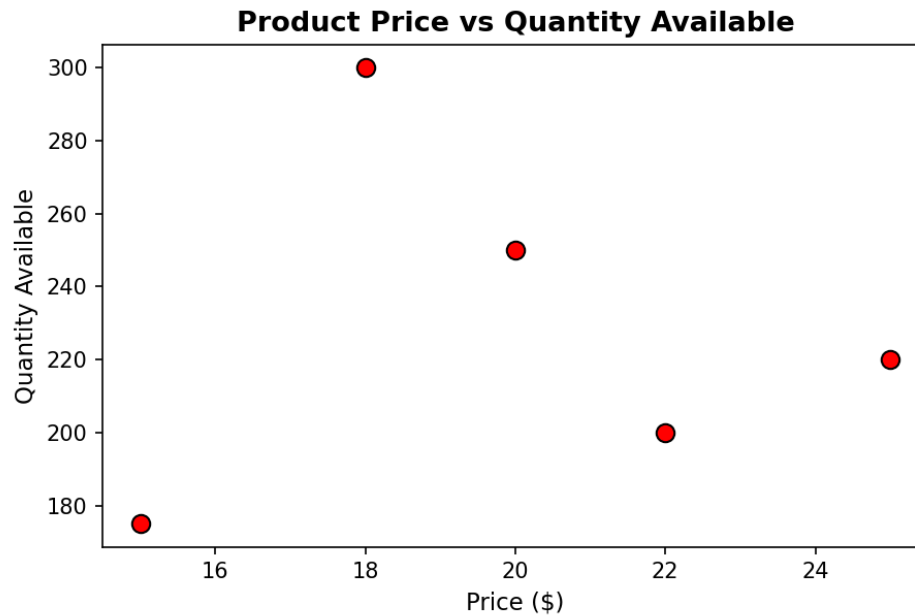
4. In R, create a scatter plot to explore the relationship between product price and quantity available. Explain any insights.

CODE:

```
# Dataset
price <- c(20, 15, 18, 22, 25)
quantity <- c(250, 175, 300, 200, 220)

# Scatter Plot
plot(price, quantity,
     pch = 19, col = "red",
     xlab = "Price ($)",
     ylab = "Quantity Available",
     main = "Product Price vs Quantity Available")
```

OUTPUT:



Insight: There is no strong or consistent linear relationship between price and quantity available in this dataset — some higher-priced products (e.g. Product D) still carry moderate stock, suggesting inventory levels are driven more by demand forecasting than by price alone.

Set 25:

Website Traffic Analysis

Dataset: Website Traffic

2. In R, generate a bar chart to show the top N days with the highest click-through rates. Label the chart elements.

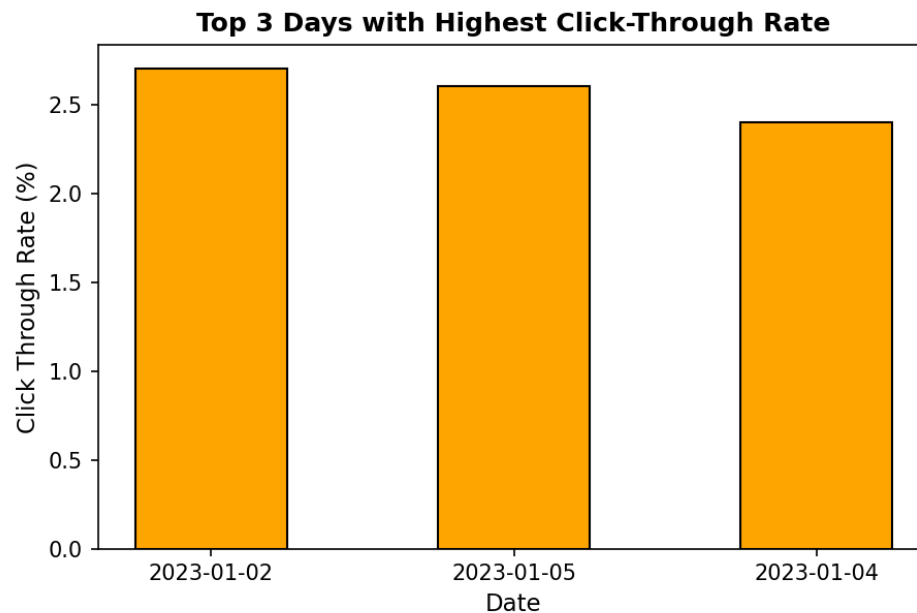
CODE:

```
# Dataset
date <- c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05")
ctr <- c(2.3, 2.7, 2.0, 2.4, 2.6)

# Select top N (N = 3) days by CTR
top_n <- 3
ordered_idx <- order(ctr, decreasing = TRUE)[1:top_n]
top_dates <- date[ordered_idx]
top_ctr <- ctr[ordered_idx]

# Bar Chart
barplot(top_ctr,
        names.arg = top_dates,
        col = "orange",
        xlab = "Date",
        ylab = "Click Through Rate (%)",
        main = "Top 3 Days with Highest Click-Through Rate")
```

OUTPUT:



3. In R, build a stacked area chart to display the distribution of user interactions (likes, shares, comments) on the website.

CODE:

```
# Dataset
date <- c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05")
likes <- c(120, 150, 100, 160, 180)
shares <- c(40, 55, 30, 50, 65)
comments <- c(20, 25, 15, 22, 28)

# Combine into matrix
interactions <- rbind(likes, shares, comments)
colnames(interactions) <- date

# Stacked Area Chart (via cumulative polygons)
x <- 1:length(date)
plot(x, colSums(interactions),
     type = "n", xaxt = "n",
     xlab = "Date", ylab = "Number of Interactions",
     main = "Stacked Area Chart of User Interactions")
axis(1, at = x, labels = date)

polygon(c(x, rev(x)), c(rep(0, length(x)), rev(likes)),
       col = "#4C72B0", border = "black")
polygon(c(x, rev(x)), c(likes, rev(likes + shares)),
       col = "#55A868", border = "black")
polygon(c(x, rev(x)), c(likes + shares, rev(likes + shares + comments)),
       col = "#C44E52", border = "black")
```

```
legend("topleft",  
      legend = c("Likes","Shares","Comments"),  
      fill = c("#4C72B0","#55A868","#C44E52"))
```

OUTPUT:

